Homework 2

1. Proof of the Fundamental Theorem of Linear Algebra

In this question, we will prove the fundamental theorem of linear algebra. For any $A \in \mathbb{R}^{m \times n}$, let $\mathcal{N}(A)$, $\mathcal{R}(A)$, and $\operatorname{rank}(A)$ denote the null space, range and rank of A respectively. For any subspace, $\mathcal{S}$ with dimension, $\dim(\mathcal{S})$, let $\mathcal{S}^\perp$ denote its the subspace orthogonal to $\mathcal{S}$. The fundamental theorem of linear algebra states that,

$$\mathcal{N}(A) \oplus \mathcal{R}\left(A^\top\right) = \mathbb{R}^n. \tag{1}$$

The proof technique we employ will first show that,

$$\mathcal{N}(A) = \mathcal{R}\left(A^\top\right)^\perp. \tag{2}$$

Then we will prove that we can find orthonormal vectors $\vec{e}_1, \vec{e}_2, \ldots \vec{e}_n$ such that $\mathcal{N}(A) = \operatorname{span}(\vec{e}_1, \vec{e}_2, \ldots, \vec{e}_\ell)$ and $\mathcal{R}\left(A^\top\right) = \operatorname{span}(\vec{e}_{\ell+1}, \vec{e}_{\ell+2}, \ldots, \vec{e}_n)$. As a corollary we get the rank-nullity theorem:

$$\dim(\mathcal{N}(A)) + \operatorname{rank}(A) = n. \tag{3}$$

(a) First, show that $\mathcal{N}(A) \subseteq \mathcal{R}\left(A^\top\right)^\perp$.

HINT: Consider $\vec{u}$ in $\mathcal{N}(A)$, $\vec{v} \in \mathcal{R}\left(A^\top\right)$ and show that $\vec{u}^\top \vec{v} = 0$.

Solution: Take a vector $\vec{u}$ from the nullspace: $A\vec{u} = 0, \vec{u} \neq 0$, and a vector $\vec{w}$ from the rowspace: $A^\top \vec{v} = \vec{w}$. Now we can show orthogonality:

$$\langle \vec{w},\ \vec{u} \rangle = \left\langle A^\top \vec{v},\ \vec{u} \right\rangle = \vec{v}^\top A\vec{u} = \vec{v}^\top \vec{0} = 0 \tag{4}$$

Therefore, $\vec{u} \in \mathcal{R}\left(A^\top\right)^\perp$. Since this holds for every u in $\mathcal{N}(A)$ we have $\mathcal{N}(A) \subseteq \mathcal{R}\left(A^\top\right)^\perp$. Note that this is not an equality since we only proved that every vector in the nullspace is orthogonal , not that these vectors are the **only** orthogonal ones.

(b) Now show that: $\mathcal{R}\left(A^\top\right)^\perp \subseteq \mathcal{N}(A)$.

HINT: Show that any vector $\vec{v}$ that is orthogonal to all vectors in the range of $A^\top$ satisfies $A\vec{v} = 0$. To do this, consider $\vec{v} \in R(A^\top)^\perp$ and what it implies for $\vec{v}^\top A^\top$.

Solution: Consider $\vec{v} \in \mathcal{R}\left(A^\top\right)^\perp$ and $\vec{u} \in \mathcal{R}\left(A^\top\right)$. They are related by $\langle \vec{v},\ \vec{u} \rangle = \vec{v}^\top \vec{u} = 0$. Each column of $A^\top$ is in $\mathcal{R}\left(A^\top\right)$. So we have $\vec{v}^\top \vec{u} = 0$ if $\vec{u}$ is a column of $A^\top$. This implies $\vec{v}^\top A^\top = \vec{0}^\top$. Taking transposes gives $(A\vec{v})^\top = \vec{0}^\top$ which implies that $A\vec{v} = \vec{0}$ and so $\vec{v} \in \mathcal{N}(A)$. Since this holds for any $\vec{v} \in \mathcal{R}\left(A^\top\right)^\perp$ we have $\mathcal{R}\left(A^\top\right)^\perp \subseteq \mathcal{N}(A)$.

(c) Note that we could apply the orthogonal decomposition theorem (theorem 19 in the course reader) at this point to complete the proof. However, instead we'll work through how to re-derive that result directly. Let $\dim(\mathcal{N}(A)) = \ell$ and let $\vec{e}_1, \ldots, \vec{e}_\ell$ be an orthonormal basis for $\mathcal{N}(A)$. Consider an extension of the basis to an orthonormal basis, $\vec{e}_1, \ldots, \vec{e}_n$ for $\mathbb{R}^n$. We will prove that $\vec{e}_{\ell+1}, \ldots, \vec{e}_n$ form a basis for $\mathcal{R}\left(A^\top\right)$ and as a consequence, the dimension of $\mathcal{R}\left(A^\top\right)$ is $n - \ell$.

i. Show that $\mathcal{R}\left(A^\top\right)$ lies in the span of $\vec{e}_{\ell+1}, \ldots, \vec{e}_n$.

HINT: Express any vector $\vec{u} \in \mathcal{R}\left(A^\top\right)$ as $\vec{u} = \sum_{i=1}^n \alpha_i \vec{e}_i$. What are the values of α_i?

Solution: We can obtain the coefficient attached to the basis vector $\vec{e}_i$ by finding the scalar projection (dot product) of $\vec{u}$ onto $\vec{e}_i$. Therefore, by projecting a vector onto an orthonormal one, we see that $\alpha_i = 0$. Now, take any $\vec{u} \in \mathcal{R}\left(A^\top\right)$. We can express it as a linear combination of A's basis vectors:

$$\vec{u} = \sum_{i=1}^{n} \alpha_i \vec{e}_i. \tag{5}$$

But we also know from parts (a) and (b) that u is orthogonal to any vectors in the nullspace, which are spanned by/include the first ℓ basis vectors:

$$\vec{u}^\top \vec{e}_i = 0 \tag{6}$$

for all $i \in \{1, 2, \ldots, \ell\}$. Therefore, $\vec{u}^\top \vec{e}_i = \alpha_i = 0$ for all $i \in \{1, 2, \ldots, \ell\}$ and subsequently

$$\vec{u} = \sum_{i=\ell+1}^{n} \alpha_i \vec{e}_i. \tag{7}$$

Therefore, any vector $\vec{u} \in \mathcal{R}\left(A^\top\right)$ can be spanned by $\vec{e}_{\ell+1}, \ldots, \vec{e}_n$, making $\mathcal{R}\left(A^\top\right)$ a subset of the span of $\vec{e}_{\ell+1}, \ldots, \vec{e}_n$.

ii. From part (i) we know that $\mathcal{R}\left(A^\top\right) \subseteq \operatorname{span}(\vec{e}_{\ell+1}, \ldots, \vec{e}_n)$, but we want something stronger. Show that in fact $\mathcal{R}\left(A^\top\right) = \operatorname{span}(\vec{e}_{\ell+1}, \ldots, \vec{e}_n)$.

HINT: First, prove $\dim\left(\mathcal{R}\left(A^\top\right)\right) = \dim(\operatorname{span}(\vec{e}_{\ell+1}, \ldots, \vec{e}_n)) = n - \ell$ *by contradiction. Assume* $\dim\left(\mathcal{R}\left(A^\top\right)\right) = k < n - \ell$*, show that a vector* $\vec{u} \in \operatorname{span}(\vec{e}_{\ell+1}, \ldots, \vec{e}_n)$ *and* $\vec{u} \notin \mathcal{R}\left(A^\top\right)$ *cannot exist.*

Specifically, let $\vec{f}_1, \vec{f}_2, \ldots, \vec{f}_k$ *be an orthonormal basis for* $\mathcal{R}\left(A^\top\right)$*, we can find non-zero* $\vec{u}_\perp = \vec{u} - \sum_{i=1}^{k} (\vec{f}_i^\top \vec{u}) \vec{f}_i$ *that is orthogonal to* $\mathcal{R}\left(A^\top\right)$*. Does* $\vec{u}_\perp$ *lie in* $\mathcal{N}(A)$*? Does* $\vec{u}_\perp$ *also lie in* $\operatorname{span}(\vec{e}_{\ell+1}, \ldots, \vec{e}_n)$*? Does this lead to a contradiction? Think of* $n - \ell = 3$ *and* $k = 2$ *for visualization.*

HINT: Second, you can use without proof the fact that for two subspaces, $\mathcal{S}_1$ *and* $\mathcal{S}_2$*, if* $\mathcal{S}_1 \subseteq \mathcal{S}_2$ *and* $\dim(\mathcal{S}_1) = \dim(\mathcal{S}_2)$ *then* $\mathcal{S}_1 = \mathcal{S}_2$*.*

Solution: Assume the contrary and let $\vec{f}_1, \ldots, \vec{f}_k$ be an orthonormal basis for $\mathcal{R}\left(A^\top\right)$. Then, there exists $\vec{u}$ in the span of $\vec{e}_{\ell+1}, \ldots, \vec{e}_n$ such that $\vec{u} \notin \mathcal{R}\left(A^\top\right)$. From this, we get $\vec{u}_\perp = \vec{u} - \sum_{i=1}^{k} (\vec{f}_i^\top \vec{u}) \vec{f}_i$ is non-zero and orthogonal to $\mathcal{R}\left(A^\top\right)$. Visually, you can think of the range of $A^\top$ as the x, y plane, and $\vec{u}$ extends into 3D space outside this plane. From there, we can define a $\vec{u}_\perp$ that is perpendicular to the x, y plane simply by using Gram-Schmidt on the basis vectors, for example. Thus, $\vec{u}_\perp \in \mathcal{R}\left(A^\top\right)^\perp = \mathcal{N}(A)$. However, we also have $\vec{u}_\perp \in \operatorname{span}(\vec{e}_{\ell+1}, \ldots, \vec{e}_n)$ which is a contradiction as $\mathcal{N}(A)$ and $\operatorname{span}(\vec{e}_{\ell+1}, \ldots, \vec{e}_n)$ are orthogonal to each other. Therefore, the dimension of $\mathcal{R}\left(A^\top\right)$ is at least $n - \ell$ and is exactly $n - \ell$ as it is contained in an $n - \ell$ dimensional space. Since, we have $\mathcal{R}\left(A^\top\right) \subseteq \operatorname{span}(\vec{e}_{\ell+1} \ldots \vec{e}_n)$, and $\dim\left(\mathcal{R}\left(A^\top\right)\right) = \dim(\operatorname{span}(\vec{e}_{\ell+1} \ldots \vec{e}_n)) = n - \ell$, we can conclude that $\mathcal{R}\left(A^\top\right) = \operatorname{span}(\vec{e}_{\ell+1}, \ldots, \vec{e}_n)$.

(d) Using part (c) argue why $\mathcal{N}(A) \oplus \mathcal{R}\left(A^\top\right) = \mathbb{R}^n$ and why the rank nullity theorem holds.

Solution: We have found a basis $\vec{e}_1, \vec{e}_2, \ldots, \vec{e}_n$ such that the first ℓ vectors $\vec{e}_1, \vec{e}_2, \ldots, \vec{e}_\ell$ form a basis for $\mathcal{N}(A)$ and the last $n - \ell$ vectors form a basis for $\mathcal{R}\left(A^\top\right)$. Thus the claim $\mathcal{N}(A) \oplus \mathcal{R}\left(A^\top\right) = \mathbb{R}^n$ follows. Also,

$$\dim(\mathcal{N}(A)) + \operatorname{rank}(A) = \dim(\mathcal{N}(A)) + \operatorname{rank}\left(A^\top\right) \tag{8}$$

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$$= \dim(\mathcal{N}(A)) + \dim\left(\mathcal{R}\left(A^\top\right)\right) \tag{9}$$

$$= \ell + n - \ell = n. \tag{10}$$

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2. Eigenvalues of Symmetric Matrices

Let $A \in \mathbb{S}^n$ (i.e., the set of $n \times n$ real symmetric matrices) with eigenvalues λ_i. Prove that all of the eigenvalues of A are real (i.e. that $\lambda_i \in \mathbb{R}$ for each i).

HINT: Consider the quantity $(Av)^* v$ for eigenvector v where $*$ denotes the conjugate transpose. Note that this is the Hermitian inner product between Av and v.

NOTE: This exercise is part of the proof of the spectral theorem.

Solution: Let $(v \in \mathbb{C}^n, \lambda \in \mathbb{C})$ be an eigen-pair of the matrix A. Consider the quantity $(Av)^* v$. We have that

$$(Av)^* v = v^* A^* v = v^* A v = \lambda v^* v$$

where the second equality uses the fact that A is symmetric and real and the third that v is an eigenvector of A with eigenvalue λ.

Similarly, we can apply the fact that $Av = \lambda v$ before expanding the inner product to get

$$(Av)^* v = (\lambda v)^* v = \overline{\lambda} v^* v$$

where $\overline{\lambda}$ notates the complex conjugate of λ. As a result,

$$\lambda v^* v = \overline{\lambda} v^* v$$

Since v is a non-zero vector, $v^* v = \|v\|_2^2 > 0$, so we can re-arrange our equality to see that

$$\lambda = \overline{\lambda}$$

The only numbers which are equal to their own complex conjugate are the real numbers, so this tells us that $\lambda \in \mathbb{R}$ as desired.

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3. Distinct Eigenvalues, Orthogonal Eigenspaces

Let $A \in \mathbb{S}^n$ (i.e. the set of $n \times n$ real symmetric matrices) and $(\lambda_1, \vec{u}_1), (\lambda_2, \vec{u}_2), \lambda_1 \neq \lambda_2$ be distinct eigen-pairs of A. Show that $\vec{u}_1^\top \vec{u}_2 = 0$, i.e., eigenspaces corresponding to distinct eigenvalues are mutually orthogonal.

HINT: First try to prove that $\lambda_1 \vec{u}_1^\top \vec{u}_2 = \lambda_2 \vec{u}_1^\top \vec{u}_2$, then show that this implies $\vec{u}_1^\top \vec{u}_2 = 0$.

NOTE: This exercise is part of the proof of the spectral theorem.

Solution: We have

$$\begin{aligned}
\lambda_1 \vec{u}_1^\top \vec{u}_2 &= (\lambda_1 \vec{u}_1)^\top \vec{u}_2 && (11)\\
&= (A\vec{u}_1)^\top \vec{u}_2 && (12)\\
&= \vec{u}_1^\top A^\top \vec{u}_2 && (13)\\
&= \vec{u}_1^\top A \vec{u}_2 && (14)\\
&= \vec{u}_1^\top (\lambda_2 \vec{u}_2) && (15)\\
&= \lambda_2 \vec{u}_1^\top \vec{u}_2. && (16)
\end{aligned}$$

Thus we have

$$\lambda_1(\vec{u}_1^\top \vec{u}_2) = \lambda_2(\vec{u}_1^\top \vec{u}_2) \implies (\lambda_1 - \lambda_2)(\vec{u}_1^\top \vec{u}_2) = 0. \tag{17}$$

Since $\lambda_1 \neq \lambda_2$, we have $\lambda_1 - \lambda_2 \neq 0$, so we must have $\vec{u}_1^\top \vec{u}_2 = 0$.

Thus, $\vec{u}_1^\top \vec{u}_2 = 0$ for any $\vec{u}_1, \vec{u}_2$ corresponding to different eigenvalues. Stated differently, unique eigenvalues correspond to orthogonal eigenvectors.

This, in combination with the fact that the geometric multiplicity and algebraic multiplicity of a symmetric matrix are equal, allows us to construct an orthonormal set of eigenvectors. First, find all the distinct eigenvalues and their respective eigenvectors. Then, for all eigenvalues with algebraic multiplicity > 1, we know that the respective eigenspace is spanned by k linearly independent eigenvectors. Utilizing Gram-Schmidt, we can construct an orthonormal set of eigenvectors from this basis for this eigenspace. Putting the eigenvectors from these two cases together, we have constructed the U matrix of the decomposition.

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4. Gram Schmidt

Any set of n linearly independent vectors in $\mathbb{R}^n$ could be used as a basis for $\mathbb{R}^n$. However, certain bases could be more suitable for certain operations than others. For example, an orthonormal basis could facilitate solving linear equations.

(a) Given a matrix $A \in \mathbb{R}^{n\times n}$, it could be represented as a multiplication of two matrices

$$A = QR, \tag{18}$$

where $Q \in \mathbb{R}^{n\times n}$ is an orthonormal matrix and $R \in \mathbb{R}^{n\times n}$ is an upper-triangular matrix. For the matrix A, describe how Gram-Schmidt process could be used to find the Q and R matrices, and apply this to

$$A = \begin{bmatrix} 3 & -3 & 1 \\ 4 & -4 & -7 \\ 0 & 3 & 3 \end{bmatrix} \tag{19}$$

to find an orthonormal matrix Q and an upper-triangular matrix R.

Solution: Let $\vec{a}_i$ and $\vec{q}_i$ denote the columns of A and Q, respectively. Using Gram-Schmidt, we obtain an orthonormal basis $\vec{q}_i$ for the column space of A.

$$\vec{p}_1 = \vec{a}_1, \vec{q}_1 = \frac{\vec{p}_1}{\|\vec{p}_1\|_2} \tag{20}$$

$$\vec{p}_2 = \vec{a}_2 - (\vec{a}_2^\top \vec{q}_1)\vec{q}_1, \quad \vec{q}_2 = \frac{\vec{p}_2}{\|\vec{p}_2\|_2} \tag{21}$$

$$\vec{p}_3 = \vec{a}_3 - (\vec{a}_3^\top \vec{q}_1)\vec{q}_1 - (\vec{a}_3^\top \vec{q}_2)\vec{q}_2, \vec{q}_3 = \frac{\vec{p}_3}{\|\vec{p}_3\|_2} \tag{22}$$

$$\vdots \tag{23}$$

Rearranging terms, we have

$$\vec{a}_1 = r_{11}\vec{q}_1 \tag{24a}$$

$$\vec{a}_i = r_{i1}\vec{q}_1 + \cdots + r_{ii}\vec{q}_i, \quad i = 2, ..., n, \tag{24b}$$

where each $\vec{q}_i$ has unit norm, and $r_{ij}\vec{q}_j$ denotes the projection of $\vec{a}_i$ onto the vector $\vec{q}_j$ for $j \neq i$. Stacking $\vec{a}_i$ horizontally into A and rewriting (24a-b) in matrix notation, we obtain $A = QR$. For the given matrix, we have

$$A = \begin{bmatrix} 0.6 & 0 & 0.8 \\ 0.8 & 0 & -0.6 \\ 0 & 1 & 0 \end{bmatrix} \begin{bmatrix} 5 & -5 & -5 \\ 0 & 3 & 3 \\ 0 & 0 & 5 \end{bmatrix}. \tag{25}$$

Note that an equivalent factorization is $A = (-Q)(-R)$.

(b) Given an invertible matrix $A \in \mathbb{R}^{n\times n}$ and an observation vector $\vec{b} \in \mathbb{R}^n$, the solution to the equality

$$A\vec{x} = \vec{b} \tag{26}$$

is given as $\vec{x} = A^{-1}\vec{b}$. For the matrix $A = QR$ from part 4(a), assume that we want to solve

$$A\vec{x} = \begin{bmatrix} 8 \\ -6 \\ 3 \end{bmatrix}. \tag{27}$$

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By using the fact that Q is an orthonormal matrix, find $\vec{v}$ such that

$$R\vec{x} = \vec{v}. \tag{28}$$

Then, given the upper-triangular matrix R in part 4(a) and $\vec{v}$, find the elements of $\vec{x}$ sequentially.

Solution: We note that $Q^{-1} = Q^\top$.

$$A\vec{x} = \vec{b} \tag{29}$$
$$QR\vec{x} = \vec{b} \tag{30}$$
$$Q^\top QR\vec{x} = R\vec{x} = Q^\top \vec{b}. \tag{31}$$

Thus

$$\vec{v} = Q^\top \vec{b} = \begin{bmatrix} 0 \\ 3 \\ 10 \end{bmatrix}. \tag{32}$$

Given R and $\vec{v}$, we can find $\vec{x}$ by back-substitution:

$$\begin{bmatrix} 5 & -5 & -5 \\ 0 & 3 & 3 \\ 0 & 0 & 5 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} 0 \\ 3 \\ 10 \end{bmatrix} \implies x_3 = 2 \implies x_2 = -1 \implies x_1 = 1 \implies \vec{x} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}. \tag{33}$$

(c) Given an invertible matrix $B \in \mathbb{R}^{n\times n}$ and an observation vector $\vec{c} \in \mathbb{R}^n$, find the computational cost of finding the solution $\vec{z}$ to the equation $B\vec{z} = \vec{c}$ by using the QR decomposition of B. Assume that Q and R matrices are available, and adding, multiplying, and dividing scalars take one unit of "computation".

As an example, computing the inner product $\vec{a}^\top \vec{b}$ is said to be $\mathcal{O}(n)$, since we have n scalar multiplication for each $a_i b_i$. Similarly, matrix vector multiplication is $\mathcal{O}(n^2)$, since matrix vector multiplication can be viewed as computing n inner products. The computational cost for inverting a matrix in $\mathbb{R}^n$ is $\mathcal{O}(n^3)$, and consequently, the cost grows rapidly as the set of equations grows in size. This is why the expression $A^{-1}\vec{b}$ is usually not computed by directly inverting the matrix A. Instead, the QR decomposition of A is exploited to decrease the computational cost.

Solution: We count the number of operations in back substitution. Solving the initial equation

$$r_{nn}x_n = \bar{b}_n \tag{34}$$

takes 1 multiplication. Solving each subsequent equation takes one more multiplication and one more addition than the previous. In total, we have $1 + 3 + 5 + \cdots$ of operations, which is on the order of $\mathcal{O}(n^2)$. Thus, matrix-vector multiplication and back substitution are both $O(n^2)$. Given the QR decomposition of A, we can solve $A\vec{x} = \vec{b}$ in $O(n^2)$ times.

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5. Determinants

Consider a unit box $\mathcal{B}$ in $\mathbb{R}^2$ — i.e., the square with corners $\begin{bmatrix}0\\0\end{bmatrix}$, $\begin{bmatrix}0\\1\end{bmatrix}$, $\begin{bmatrix}1\\0\end{bmatrix}$, and $\begin{bmatrix}1\\1\end{bmatrix}$. Define $A(\mathcal{B})$ as the parallelogram generated by applying matrix A to every point in $\mathcal{B}$.

(a) For $A = \begin{bmatrix}1 & 2\\3 & 4\end{bmatrix}$, calculate the location of each corner of $A(\mathcal{B})$.

Solution: Directly applying matrix A to each corner point, we have

$$A\begin{bmatrix}0\\0\end{bmatrix} = \begin{bmatrix}1 & 2\\3 & 4\end{bmatrix}\begin{bmatrix}0\\0\end{bmatrix} = \begin{bmatrix}0\\0\end{bmatrix}.$$
$$A\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}1 & 2\\3 & 4\end{bmatrix}\begin{bmatrix}0\\1\end{bmatrix} = \begin{bmatrix}2\\4\end{bmatrix}.$$
$$A\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}1 & 2\\3 & 4\end{bmatrix}\begin{bmatrix}1\\0\end{bmatrix} = \begin{bmatrix}1\\3\end{bmatrix}.$$
$$A\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}1 & 2\\3 & 4\end{bmatrix}\begin{bmatrix}1\\1\end{bmatrix} = \begin{bmatrix}3\\7\end{bmatrix}.$$

(b) Write the area of $A(\mathcal{B})$ as a function of $\det(A)$.

HINT: How are the basis vectors $\begin{bmatrix}0\\1\end{bmatrix}$ *and* $\begin{bmatrix}1\\0\end{bmatrix}$ *transformed by the matrix multiplication?*

Solution: The area of the parallelogram $A(\mathcal{B})$ can be computed geometrically from its corner points and is equal to $|\det(A)|$. A full explanation of this relation can be found in Calafiore & El Ghaoui section 3.3.2.

(c) Calculate the area of $A(\mathcal{B})$ for each of the following values of A.

i. $A = \begin{bmatrix}1 & 2\\3 & 4\end{bmatrix}$

ii. $A = \begin{bmatrix}2 & 1\\4 & 3\end{bmatrix}$

iii. $A = \begin{bmatrix}1 & 2\\2 & 4\end{bmatrix}$

iv. $A = \begin{bmatrix}0 & -1\\1 & 0\end{bmatrix}$

Solution: We use the result from part 5(b) and use the determinant of A to calculate the area.

i. $\det(A) = \det\left(\begin{bmatrix}1 & 2\\3 & 4\end{bmatrix}\right) = -2 \implies \text{area}[A(\mathcal{B})] = |\det(A)| = 2.$

ii. $\det(A) = \det\left(\begin{bmatrix}2 & 1\\4 & 3\end{bmatrix}\right) = 2 \implies \text{area}[A(\mathcal{B})] = |\det(A)| = 2.$

iii. $\det(A) = \det\left(\begin{bmatrix}1 & 2\\2 & 4\end{bmatrix}\right) = 0 \implies \text{area}[A(\mathcal{B})] = |\det(A)| = 0$ (Singular matrix!).

iv. $\det(A) = \det\left(\begin{bmatrix}0 & -1\\1 & 0\end{bmatrix}\right) = 1 \implies \text{area}[A(\mathcal{B})] = |\det(A)| = 1$ (Rotation matrix!).

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6. Homework Process

With whom did you work on this homework? List the names and SIDs of your group members.

NOTE: If you didn't work with anyone, you can put "none" as your answer.

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